Comprehensive Literature Review

Deep Learning & Explainable AI (XAI) in Breast Cancer Imaging

ARTICLE TITLE	AUTHOR	WORK DONE	IDENTIFIED RESEARCH GAP
An Explainable Deep Learning Framework for Breast Cancer Classification Using EfficientNetV2-B0 and Grad-CAM	Abdullah et al.	Employed EfficientNetV2-B0 for classification combined with Grad-CAM to provide visual explanations of the model's decision-making process.	Primarily classification-oriented; lacks automated segmentat to isolate lesions from background noise.
Automatic classification of ultrasound breast lesions using a deep convolutional neural network mimicking human decision-making	Yap et al.	Developed a DCNN architecture designed to replicate the heuristic decision-making patterns used by clinical radiologists.	Model interpretability is less granular compared to modern X techniques; limited by the computational constraints of its tim
Revolutionizing Breast Ultrasound Diagnostics with EfficientNet-B7 and Explainable AI	Khan et al.	Utilized the heavy EfficientNet-B7 architecture to achieve high diagnostic accuracy, supported by Grad-CAM for localization.	High computational overhead makes it difficult to deploy in lo resource clinical settings; absence of a segmentation stage.
A Dual-stage Deep Learning Framework for Breast Ultrasound Image Segmentation and Classification	Bruno et al.	Proposed a two-step pipeline that first performs lesion segmentation followed by precise ROI-based classification.	Increased pipeline complexity; error propagation from the segmentation stage can negatively impact final classification
Deep learning radiomics on grayscale ultrasound images assists in diagnosing benign and malignant of BI-RADS 4 lesions	Scientific Reports	Integrated deep learning with radiomic feature extraction to differentiate between benign and malignant BI-RADS 4 category lesions.	Niche focus on BI-RADS 4; may not generalize effectively to other categories or multi-modal imaging data.
Advancing Breast Cancer-AI Diagnostics: An Explainable Deep Learning Model Using 2D Grayscale Ultrasound Imaging	Ahmed et al.	MobileNetV2-based model for 3-class classification using BUSI dataset with Grad-CAM explainability.	Limited generalization due to single-dataset training; basic explainability lacks deep clinical correlation.
An Explainable AI-driven Deep Neural Network for Accurate Breast Cancer Detection from Histopathological and Ultrasound Images	Zhang et al.	DenseNet121-based multi-modal model (ultrasound + histopathology) achieving high accuracy (~93%).	Lacks real-time segmentation; high computational cost for m modal processing; limited real-world clinical validation.

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Explainable AI for Breast Cancer Diagnosis (MDPI)	Li et al.	Applied SHAP and Grad-CAM to interpret deep learning predictions, focusing on feature importance.	Lack of seamless integration with segmentation; high latency generating SHAP values for high-resolution images.
Breast Cancer Detection using Machine Learning Techniques (PubMed)	Cruz et al.	Applied traditional ML (SVM, Random Forest) to breast cancer feature datasets.	Significantly lower accuracy than DL; cannot process raw pix data without manual feature engineering; no explainability.